SKILLNAVORA · AI PRODUCT MANAGEMENT & GENAI LAUNCHPAD

Multimodal AI & AI Media

Use AI that sees, hears and creates: vision understanding, image generation, video and voice, multimodal product use cases, and rights, safety, deepfakes and content labelling.

Edition 2026 · skillnavora.in

How to use this guide

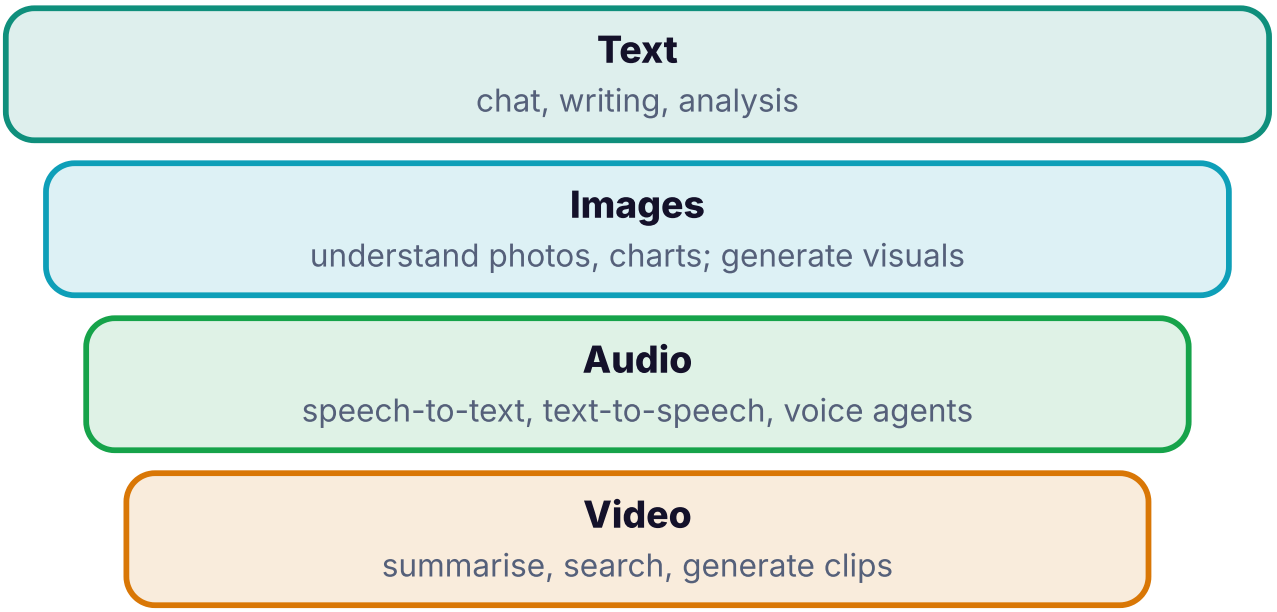
Each class has goals, concepts with Indian examples, key takeaways and a practice exercise, from basic to advanced. Finish with the final project to build something you can show in interviews.

	Class	Level
Class 1	**Multimodal basics**	Basic
Class 2	**Vision understanding use cases**	Basic
Class 3	**How multimodal AI works (without the maths)**	Basic
Class 4	**Image generation in products**	Intermediate
Class 5	**Voice and video**	Intermediate
Class 6	**Speech and voice: ASR, TTS and voice agents**	Intermediate
Class 7	**Image and video generation for products and marketing**	Intermediate
Class 8	**Rights, safety, deepfakes and labelling**	Advanced
Class 9	**Multimodal search, RAG and recommendations**	Advanced
Class 10	**Evaluating multimodal quality, safety and cost**	Advanced
Class 11	**Law, ethics and provenance for AI media in India and globally**	Advanced
Class 12	**Case studies: three multimodal features**	Advanced
Class 13	**Interview practice: multimodal scenario questions**	Advanced
Class 14	**Final project**	Advanced
	Toolkit: templates, cheat sheet, glossary, 13 interview questions, further reading	

BASIC · CLASS 1

Multimodal basics

Know what modern models can see, hear and create.



Modalities

Modalities at a glance

Modality	Understanding (input)	Generation (output)	Typical products
Text	Summaries, classification, extraction	Writing, code, chat	Assistants, support, content
Images	OCR, object detection, visual Q&A	Image creation and editing	Product photos, ads, document AI
Audio / speech	Speech-to-text, speaker and emotion detection	Text-to-speech, music, voice cloning	Voice assistants, call analytics, dubbing
Video	Scene search, summaries, safety checks	Clips, avatars, animation	Explainers, ads, learning, surveillance review
Documents and tables	Layout understanding, extraction	Reports, slides	Invoices, KYC, contracts

Key takeaways

- Models now take and produce many modalities.
- Choose the modality users already use.
- Voice matters in India (literacy, languages).

Practice exercise: List 5 features where a photo or voice input beats typing.

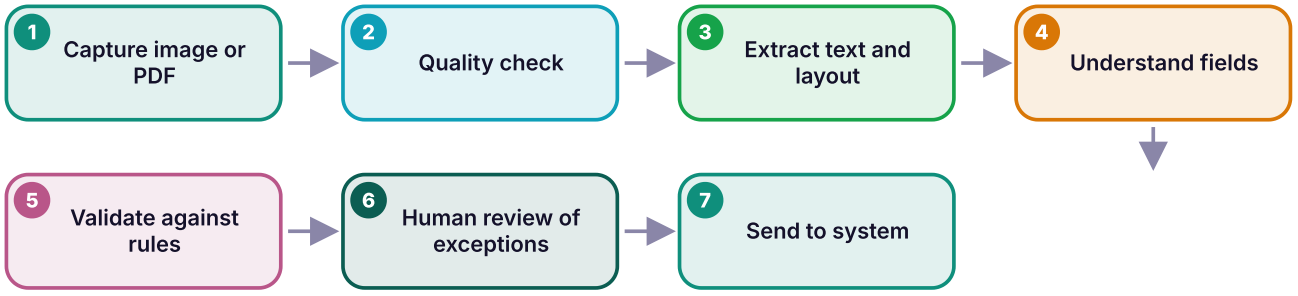
BASIC · CLASS 2

Vision understanding use cases

Use AI to read and interpret images.

- OCR of invoices, IDs (with consent), forms; handwritten notes.
- Chart and screenshot understanding; UI review.
- Quality inspection (manufacturing), crop disease, damage assessment (insurance).
- Accessibility: describing images for visually impaired users.

Document AI pipeline



From photo to structured data

- Measure **field-level accuracy** (invoice number, date, amount), not just overall text accuracy.
- Use **confidence thresholds**: auto-accept high confidence; route low confidence to people.
- Cross-check totals and dates with business rules (tax = rate x amount).

Key takeaways

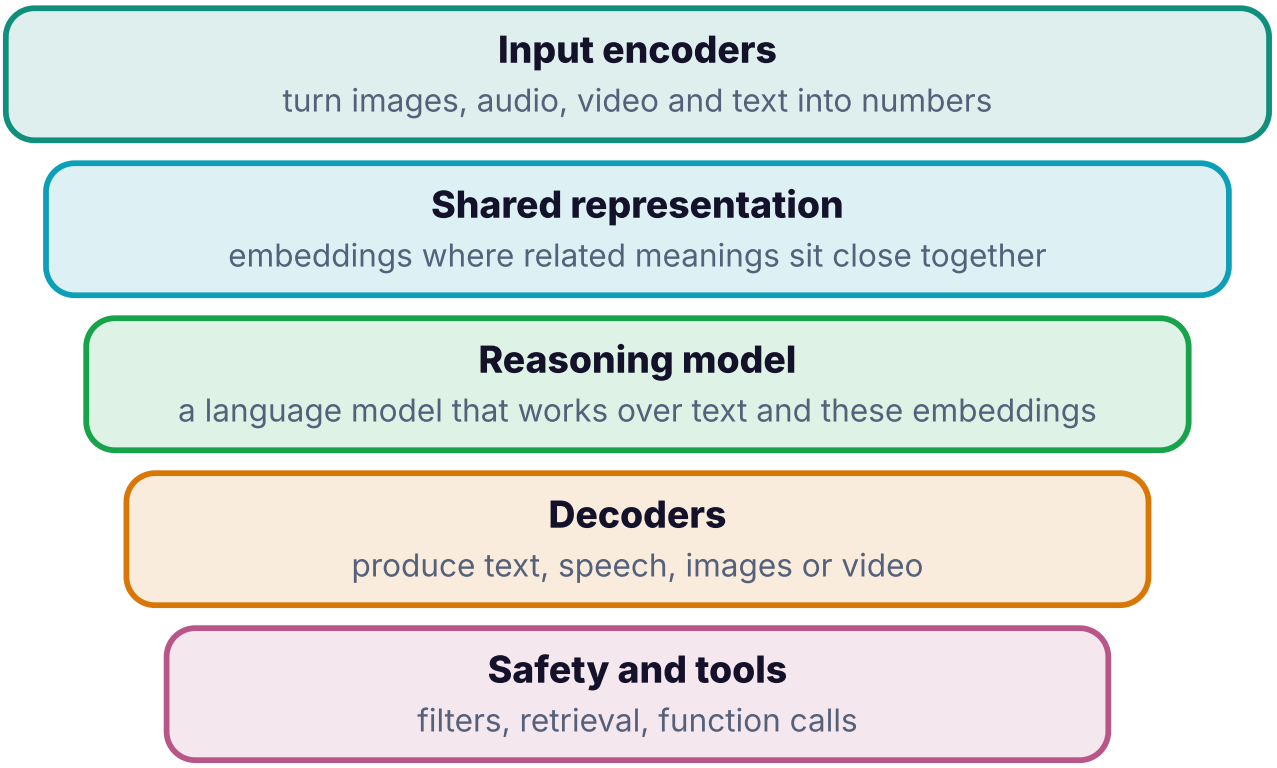
- Vision unlocks offline-to-digital workflows.
- Mask personal data.
- Measure accuracy on your own images.

Practice exercise: Design an AI feature that reads shop receipts for expense tracking.

BASIC · CLASS 3

How multimodal AI works (without the maths)

Understand the building blocks so you can reason about capabilities and limits.



A simplified multimodal system

- **Embeddings:** an image of a red kurta and the words 'red kurta' map to nearby points. Models such as CLIP learn this alignment, which powers image search.
- **Vision-language models:** read images and answer questions about them; strong at charts and documents, weaker at precise counting or tiny details.
- **Diffusion models:** generate images by starting with noise and removing it step by step following a text prompt.
- **Speech models:** speech-to-text (ASR) turns audio into text; text-to-speech (TTS) turns text into audio; voice cloning copies a voice from a sample.
- **Video:** treated as sequences of frames plus audio; generation is expensive and improving fast.

Limits to remember: models can misread blurry images, miss small text, hallucinate details, mishandle unusual accents, and show bias. Always test on your own data.

Key takeaways

- Embeddings link different modalities.
- Vision models understand; diffusion models generate.
- Test limits on your own data.

Practice exercise: Explain to a colleague, in 5 sentences, how a photo search for 'blue formal shirt' works.

INTERMEDIATE · CLASS 4

Image generation in products

Use generative images responsibly.

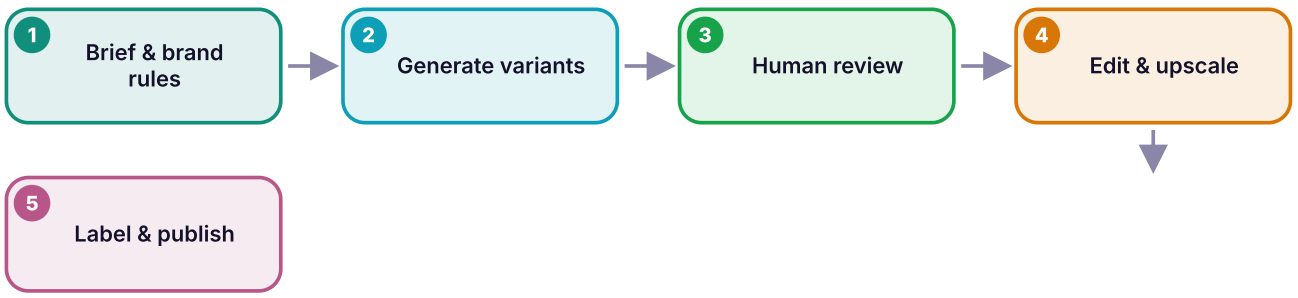


Image generation workflow

Tools: Midjourney, Adobe Firefly, Ideogram, Leonardo, DALL·E, Imagen, Canva AI. Use cases: ads, product mock-ups, thumbnails, illustrations.

Image generation workflow for a retail catalogue

Step	Detail
Brief	Product, angle, background, lighting, brand colours, what must not change
Generate	Use product photos as reference to keep the real item accurate
Review	Check logos, text, hands, proportions; reject images that misrepresent the product
Edit	Inpainting, background swap, resizing
Label and publish	Disclose AI-generated imagery where policy or law requires; keep source files

Key takeaways

- Keep brand rules in the brief.
- Human review before publishing.
- Prefer tools with clear commercial-use terms.

Practice exercise: Create a brand-safe image brief template.

INTERMEDIATE · CLASS 5

Voice and video

Build with speech and video AI.

- Speech-to-text and text-to-speech in many Indian languages; voice agents (see Voice Agents lesson).
- Video: AI summaries and search of recordings; avatar and generative video tools (Runway, Veo, Sora, HeyGen, Synthesia) for explainers.
- Latency, accents, background noise and cost are key constraints.

Voice agent latency budget (example)

Stage	Target
Speech detection and streaming to server	100 to 200 ms
Speech to text	200 to 300 ms
Language model response (first tokens)	300 to 600 ms
Text to speech (first audio)	150 to 300 ms
Total perceived delay	Under about 1 second feels natural

Key takeaways

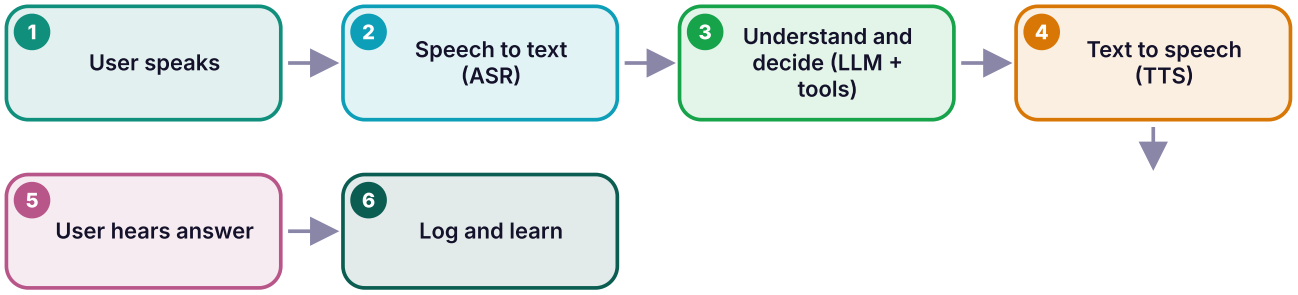
- Test with real accents and noise.
- Video AI is great for explainers and summaries.
- Watch latency and cost.

Practice exercise: Plan a Hindi voice FAQ assistant for a clinic.

INTERMEDIATE · CLASS 6

Speech and voice: ASR, TTS and voice agents

Build voice experiences for call centres, apps and devices.



A voice assistant pipeline

Topic	What PMs should know
ASR quality	Measured by word error rate (WER). Test with real accents, noise and phone audio, and Indian names and addresses.
Streaming and latency	Stream audio and responses; allow interruption (barge-in); keep total delay near one second.
Languages	Support Hindi, Tamil, Telugu, Bengali and code-mixed speech where your users are; test each.
TTS quality	Natural, clear, appropriate pace; choose consented voices.
Telephony	Integrate with call systems; handle DTMF (keypad) input and call transfer.
Use cases	Order status, appointment booking, collections reminders, lead qualification, interview practice, call summarisation and QA.
Compliance	Call recording consent, DND rules, data retention, disclosure that the caller is an AI.
Costs	Per-minute ASR and TTS plus LLM tokens; budget cost per call.

See also the Voice Agents module in the PM Master Handbook for a step-by-step build.

Key takeaways

- Measure WER on real audio.
- Latency and interruption make voice feel human.
- Disclose AI, get consent and respect call rules.

Practice exercise: Design a voice assistant for appointment booking in Hindi and English, with the cost per call.

INTERMEDIATE · CLASS 7

Image and video generation for products and marketing

Use generative media for catalogues, ads and learning with quality control.

Use case	How it works	Quality checks	Risk
Product photography	Place real product images into generated scenes or backgrounds	Colour accuracy, shape, labels, no added features	Misleading buyers; returns
Ad creatives	Generate variants for different audiences and sizes	Brand rules, legal claims, A/B test	Copyright, platform ad rules
Virtual try-on	Overlay garments or accessories on user photos	Fit realism, diversity of bodies	Privacy of user photos
Explainer videos	Script to AI voice and visuals or avatar	Accuracy, tone, captions	Deepfake perception, disclosure
Personalised content	Insert name, city, offer into media	Data protection, consent	Creepiness
Design ideation	Rapid concept images	Not final assets without review	Originality

- **Control tools:** reference images, inpainting, outpainting, masks, style presets and brand templates keep outputs on-brand.
- **Evaluate:** human review rubrics (accuracy, brand fit, safety), automated checks (text legibility, banned content), and business metrics (CTR, conversion, return rate).
- **Cost:** per image or per second of video; cache and reuse; generate in batches.

Key takeaways

- Keep the real product accurate.
- Use brand and control tools.
- Review and A/B test before scaling.

Practice exercise: Create a workflow and review rubric for generating product images for 500 SKUs.

ADVANCED · CLASS 8

Rights, safety, deepfakes and labelling

Manage the risks of AI media.

- Copyright and licences: check tool terms for commercial use; avoid copying artists or brands.
- Consent for faces and voices; never clone voices without permission.
- Label AI-generated media; content credentials (C2PA) and watermarking.
- Moderation of user-generated AI media; follow Indian advisories on deepfakes.

Risk map for AI media

Risk	Example	Control
Copyright	Output resembles a protected work	Licensed tools, reference checks, avoid 'in the style of living artist' prompts
Likeness and voice rights	Cloned celebrity voice in an ad	Written consent, contracts, blocklists
Deepfakes and fraud	Fake CEO video asks for payment	Verification procedures, provenance, detection
Misinformation	Fake news images	Labelling, review, platform policy
Privacy	Faces and voices are biometric-like data	Consent, minimisation, retention limits
Bias	Stereotyped generations	Prompt guidelines, review, testing

Key takeaways

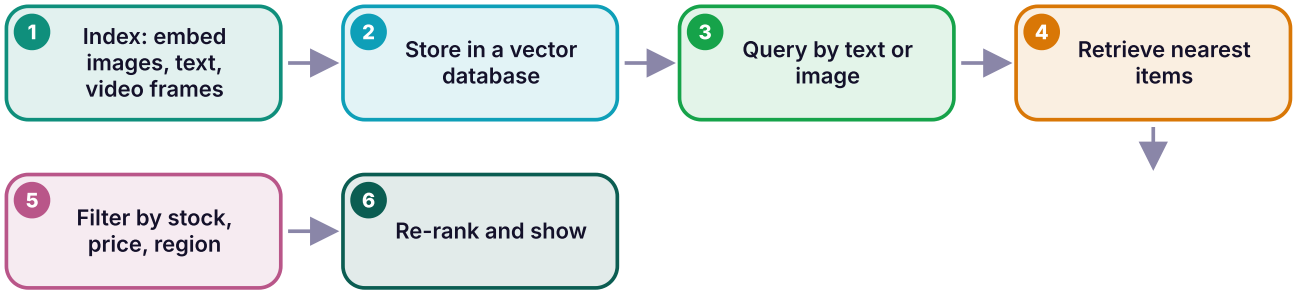
- Consent for likeness and voice.
- Label AI media.
- Check commercial-use rights.

Practice exercise: Write an AI media policy for a marketing team.

ADVANCED · CLASS 9

Multimodal search, RAG and recommendations

Search across text, images and video.



Visual and semantic search pipeline

- **Visual search:** snap a photo, find similar products ("shop the look").
- **Video search:** find the moment in a lecture or meeting where a topic is discussed (transcribe, split into segments, embed).
- **Multimodal RAG:** answer questions using documents with images and tables (charts, scanned forms) by extracting text and also keeping images for vision models.
- **Hybrid ranking:** combine similarity with business rules (stock, margin, delivery).
- **Evaluation:** relevance judgments by humans, click-through and conversion, zero-result rate, latency.

Challenge	Mitigation
Embedding drift when models change	Re-index and version embeddings
Costly indexing of large media	Sample frames, batch processing, tiered storage
Privacy of faces	Avoid face recognition unless lawful and consented
Bias in visual similarity	Audit results across skin tones, body types, regions

Key takeaways

- Embed everything into one searchable space.
- Combine similarity with business rules.
- Audit for privacy and bias.

Practice exercise: Design a 'shop the look' feature, its pipeline and its metrics.

ADVANCED · CLASS 10

Evaluating multimodal quality, safety and cost

Measure what matters before and after launch.

Modality	Quality metrics	Safety checks
OCR / document AI	Field accuracy, exact-match rate, straight-through rate	PII masking, access control
Vision Q&A	Accuracy on labelled images, hallucination rate	Unsafe image detection, privacy blur
Speech	WER, intent success, call completion	Consent, DND, abusive-content handling
Image generation	Prompt adherence, brand score, human preference	NSFW, copyright, likeness filters
Video	Realism, sync, user comprehension	Deepfake policy, labelling

- **Golden sets** for each modality with edge cases (blurry, low light, accents, noise, tilted forms).
- **Latency and cost** budgets per request; use smaller models and caching where possible.
- **Monitoring:** sample outputs weekly, track drift (new camera types, new accents), track complaints and escalations.

Key takeaways

- Build golden sets per modality.
- Budget latency and cost.
- Monitor drift and complaints.

Practice exercise: Create an evaluation plan for an invoice-reading feature with thresholds.

ADVANCED · CLASS 11

Law, ethics and provenance for AI media in India and globally

Handle rights, consent, disclosure and deepfakes responsibly.

- **Copyright:** rules on training data and outputs are evolving and differ by country; use tools with clear commercial terms, keep records of prompts and sources, avoid imitating specific artists or brands.
- **Personality and likeness rights:** faces and voices of real people need consent; courts in India have protected celebrity personality rights against unauthorised AI use.
- **Deepfake rules:** India's IT Rules and government advisories require platforms to act on deepfakes and encourage labelling of synthetic content; check current requirements.
- **Data protection:** biometrics-like data (faces, voiceprints) need consent, purpose limitation and strong security under the DPDP Act.
- **Provenance:** content credentials (C2PA) and watermarks help show where media came from; add visible labels where appropriate.
- **Advertising rules:** disclose AI use where it could mislead; follow ASCI guidance.

Policy item	Example rule
Disclosure	Label AI-generated images and videos in marketing and news-like contexts
Consent	Written consent for any real person's face or voice
Prohibited	Sexual, deceptive political, impersonation and child-safety violations
Review	Human review before publishing sensitive categories
Incident	Takedown process and owner within 24 hours

> Not legal advice: consult counsel for your markets, and review rules regularly.

Key takeaways

- Get consent for faces and voices.
- Label synthetic media and keep provenance.
- Write a clear AI media policy with owners.

Practice exercise: Draft an AI media policy page for your team with five rules and an incident process.

ADVANCED · CLASS 12

Case studies: three multimodal features

Walk through realistic product designs (illustrative).

Case 1: Expense receipts from photos

1. **Flow:** photo → quality check → OCR + AI extraction (vendor, date, amount, tax) → validation → user confirms → expense created.
2. **Accuracy plan:** test 500 real receipts; target 95% field accuracy on amount and date; below threshold goes to review.
3. **Cost:** Rs 0.40 per receipt vs Rs 12 manual entry.
4. **Risks:** blurry photos, handwritten receipts, duplicates (use hash and amount-date check).

Case 2: Hindi voice assistant for appointment booking

1. **Pipeline:** telephony → ASR (Hindi/English) → intent and slots → availability API → TTS confirmation → SMS.
2. **Targets:** intent success 85%, latency under 1.2 seconds, escalation to a human below 70% confidence.
3. **Compliance:** disclose AI, record with consent, respect DND.

Case 3: AI product images for a fashion catalogue

1. Use real garment photos on generated backgrounds; keep colour and fit accurate.
2. Human review of 10% sample plus automated colour-difference checks; A/B test conversion and returns.
3. Disclose AI imagery; keep source files; avoid generating models that look like real people without consent.

Key takeaways

- Define accuracy per field and per language.
- Plan human review for low confidence.
- Test business impact, including returns.

Practice exercise: Choose a case and write targets, risks and monitoring.

ADVANCED · CLASS 13

Interview practice: multimodal scenario questions

Answer multimodal product questions.

Q. How would you build a feature that reads Aadhaar or PAN cards from photos?
Model answer: I would treat it as sensitive: get consent, mask data, use secure storage, run quality checks, extract with OCR and validate formats and checksums, route low confidence to manual review, minimise retention, and comply with KYC rules. I would measure field accuracy, completion rate and time.
Why it works: Privacy-first design.

Q. How do you evaluate a voice assistant?
Model answer: Use word error rate on real audio, intent success, task completion, latency, escalation and satisfaction, tested across accents and noise.
Why it works: Complete metrics.

Q. What are the risks of AI-generated product photos?
Model answer: Misrepresenting the product, brand and copyright issues, bias, regulation and customer trust; mitigate with real references, reviews, labels and return-rate monitoring.
Why it works: Risk and mitigation.

Q. How do you reduce the cost of video features?
Model answer: Sample frames, transcribe once and reuse, cache, use smaller models, batch processing, limit resolution and length, and tier by value.
Why it works: Cost levers.

Key takeaways

- Treat sensitive images and voices carefully.
- Measure per modality.
- Control costs with sampling and caching.

Practice exercise: Write your answer to the KYC photo question with a compliance checklist.

ADVANCED · CLASS 14

Final project

Ship a multimodal feature.

Design a multimodal feature (photo receipts, voice ordering, or video course summaries): user flow, model/tool choice, accuracy eval on 30 samples, privacy and consent, labelling, cost estimate.

Key takeaways

- Evaluate on real samples.
- Design for consent and privacy.
- Estimate cost per use.

Practice exercise: Complete the multimodal project.

TOOLKIT

Templates

AI media brief

Field	Fill in
Purpose & audience	
Brand rules	
Tool & licence	
Consent needed?	
Review steps	
Labelling	
Accessibility (alt text, captions)	

TOOLKIT

Cheat sheet

Term / tool	Meaning / use
OCR	Reading text from images
TTS / STT	Text-to-speech / speech-to-text
C2PA	Content credentials standard for media provenance
Deepfake	Realistic fake media of real people
Document AI	Capture, extract, validate, review exceptions
Voice latency	Aim under about 1 second
Media risks	Copyright, likeness, deepfake, privacy, bias
Search	Embed, index, retrieve, filter, re-rank

Glossary

Watermark	Hidden or visible mark identifying AI content
ASR	Automatic speech recognition
TTS	Text-to-speech
WER	Word error rate
OCR	Optical character recognition
CLIP	A model that aligns images and text in one embedding space
Diffusion model	Image generator that denoises random noise
Inpainting	Editing part of an image
VLM	Vision-language model
Barge-in	User interrupting voice output
Provenance	Record of origin of media
Straight-through processing	Documents processed with no human touch

INTERVIEW PREP

13 interview questions with answers

Q1. What is multimodal AI?
Models that understand and/or generate several types of data: text, images, audio and video.

Q2. A good vision use case?
Reading receipts or forms from photos to remove manual entry, with accuracy checks and privacy.

Q3. Risks of AI media?
Copyright, consent for likeness and voice, deepfakes, misinformation; mitigate with consent, labelling, C2PA and moderation.

Q4. Why voice in India?
Many users prefer speaking, many languages and literacy levels; voice widens access.

Q5. What is multimodal AI?
AI that can process or generate more than one type of data, such as text, images, audio and video, often in a single model.

Q6. Where does vision AI add the most business value?
Where manual reading of images and documents is frequent: invoices, KYC, forms, quality inspection and claims, with confidence thresholds and human review for exceptions.

Q7. How do you evaluate a document AI feature?
Field-level accuracy on a labelled set, straight-through processing rate, error types, cost per document and time saved, with edge cases like blurry photos and handwriting.

Q8. What is word error rate?
A speech recognition metric: the share of words that are wrong (substitutions, deletions, insertions) compared with a reference transcript.

Q9. What makes voice agents feel natural?
Low latency (about one second), interruption handling, clear voices, accurate recognition of accents and names, and graceful recovery.

Q10. What are the main risks of AI-generated media?
Copyright, likeness and consent, deepfakes and fraud, misinformation, privacy and bias; mitigate with consent, labelling, provenance and review.

Q11. How does visual search work?
Images are embedded into vectors, stored in a vector database, and nearest vectors to a query image or text are retrieved and re-ranked with business rules.

Q12. How would you control the cost of multimodal features?
Use smaller models, cache results, batch processing, sample video frames, set budgets per request and monitor cost per outcome.

Q13. How do you handle consent for voice cloning?
Obtain explicit written consent, define scope and duration, secure samples, allow withdrawal, and prohibit impersonation.

LEARN MORE

Further reading and tools

- C2PA content credentials: https://c2pa.org/
- Google AI Studio (Gemini multimodal): https://aistudio.google.com/
- Adobe Firefly: https://www.adobe.com/products/firefly.html
- ElevenLabs docs: https://elevenlabs.io/docs
- Runway Academy: https://academy.runwayml.com/
- OpenAI CLIP paper and docs: https://openai.com/research/clip
- C2PA content credentials: https://c2pa.org/
- Hugging Face: audio course: https://huggingface.co/learn/audio-course
- ASCI guidelines: https://www.ascionline.in/

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